

Name: _____

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MyDesign Portfolio

Element B Initial Template

How to use this template

Element B: Documentation and Analysis of Prior Design Attempts

Ionization Smoke Detector



Details

Ionization smoke detectors detect fast-flaming fires by sensing small particles of smoke in the air (NFPA). When smoke enters the chamber, it disrupts an electric current and triggers a loud alarm. These devices are small and can be installed almost anywhere in a building. According to the CPSC, smoke alarms significantly reduce the risk of death in home fires.

This device is successful at detecting smoke and alerting people inside the building. However, it does not put out the fire. If someone is trapped or cannot hear the alarm, the fire may continue spreading. Information from the National Fire Protection Association (NFPA) and U.S. Consumer Product Safety Commission (CPSC).

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Strobe Light Fire Alarm System



Details

Strobe fire alarms use bright flashing lights to notify people of a fire emergency. They are especially important in public buildings to assist individuals who are deaf or hard of hearing (NFPA). The ADA requires visual alarm signals in many public facilities to improve safety.

Analysis

This device helps people who cannot hear alarms by using bright lights. However, it does not put out the fire. It also may not help people who cannot see the flashing lights. Information from National Fire Protection Association (NFPA) and Americans with Disabilities Act (ADA) accessibility guidelines.

Manual Pull Station Fire Alarm



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A manual pull station allows a person to activate the fire alarm system by pulling a lever. Many systems include a protective glass cover to prevent false alarms. When activated, it alerts everyone in the building.

Analysis

This device only alerts people and does not put out the fire. If no one pulls the alarm, it will not activate. According to Kimble Company article on fire alarm installed failures the device can also malfunction if water gets on it. The glass cover may also slow down response time in an emergency. Information from National Fire Protection Association (NFPA) fire alarm code standards.

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Automatic Fire Sprinkler System



Details

Automatic fire sprinkler systems detect heat from a fire and release water directly onto the flames. According to FEMA, sprinklers can control or extinguish fires before firefighters arrive. However, sprinklers require proper installation and connection to a water supply system.

Analysis

This device is effective at putting out or controlling fires. However, it does not always alert people loudly unless connected to an alarm system. It also depends on proper installation and water pressure to work correctly. Information from National Fire Protection Association and Federal Emergency Management Agency (FEMA).

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Battery power smoke alarm system



Details

Battery-powered smoke alarms detect smoke and sound an alarm when fire is present. They are easy to install and commonly used in homes. The U.S. Fire Administration recommends checking batteries regularly to ensure they work properly.

Analysis

This device only detects smoke and does not put out the fire. It also depends on working batteries. If the batteries die and are not replaced, the alarm may fail during a fire, which can lead to serious injuries. Information from U.S. Fire Administration and National Fire Protection Association.

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Sources: [Fire alarm system - Wikipedia](#)[10 Steps On How To Test Fire Detectors - Fire Safety Alarms](#)<https://www.phscompliance.co.uk/images/fire-alarm-box.jpg/rs-1200x675a.jpg>[32f2e82d6db4f1f42a032ebd4883ed8f.png \(1080×1080\)](#)[smoke-detector-battery.jpeg \(1200×800\)](#)